Homework — 1.4.1 Data Types — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] 156 = 128 + 16 + 8 + 4. Place values: 128(1) 64(0) 32(0) 16(1) 8(1) 4(1) 2(0) 1(0). Check: 128 + 16 + 8 + 4 = 156. ✓ Answer: **`10011100`**. *Mark: valid method — place values used or repeated division by 2 (1); correct 8-bit answer `10011100` (1).*

2. [2] Split into nibbles from the right: `1100 1010`. `1100` = 8 + 4 = 12 = **C**; `1010` = 8 + 2 = 10 = **A**. Answer: **`CA`**. *Mark: correct nibble split with values 12 and 10 (1); correct hexadecimal `CA` (1).*

3. [2] 6 = 6, F = 15. (6 × 16) + 15 = 96 + 15 = **111**. *Mark: 6 × 16 = 96 (1); + 15 = 111 (1).*

4. [4] 52 − 37 = 52 + (−37). +52 = `00110100` (32 + 16 + 4 = 52). ✓ +37 = `00100101` (32 + 4 + 1 = 37). ✓ −37: invert `00100101` → `11011010`, then + 1 → `11011011`. Add:

```
  00110100   (52)
+ 11011011   (−37)
-----------
1 00001111
```

The carry out of the most significant bit is **discarded** (the ninth bit is dropped), leaving `00001111`. `00001111` = 8 + 4 + 2 + 1 = **15**. ✓ (52 − 37 = 15) *Mark: +52 and +37 correct in binary (1); correct two's complement of 37 = `11011011` (1); correct binary addition giving `1 00001111` (1); states the final carry is discarded, giving `00001111` = 15 (1).*

5. [2] The sign bit is **0**, so 0s are copied in from the left; bits move right two places and the two rightmost bits are lost: `01001100` → `00100110` → `00010011`. Answer: **`00010011`**. Effect: division by 2^2 = **÷ 4** (integer division). Check: `01001100` = 76; `00010011` = 16 + 2 + 1 = 19 = 76 ÷ 4. ✓ *Mark: correct shifted pattern `00010011` (1); states effect = ÷4 / ÷2^2 (1).*

6. [2] Integers are stored exactly as whole binary numbers, whereas real numbers are stored in floating point as a mantissa and exponent, which can only *approximate* many values (1). So storing the score as a real risks rounding/representation errors (e.g. comparisons failing / the stored value not being exactly the score), and it typically wastes storage and is slower to process — an integer stores every whole-number score exactly (1). *Max 2: one representational point plus one consequence.*

Part B (6 marks)

7. [1] The MAR holds the **address** of the memory location about to be read from or written to — during fetch, the address of the next instruction copied from the PC (1).

8. [3] Virtual memory uses an area of **secondary storage** (a page/swap file) as if it were an extension of RAM (1). When RAM is full, pages of a program not currently in use are moved out to disk, and moved back into RAM when they are needed (1). This allows a program larger than physical RAM to run, although accessing swapped-out pages is much slower than RAM (excessive swapping = disk thrashing) (1).

9. [2] Difference (1): lossless compression allows the original file to be reconstructed *exactly*, whereas lossy compression permanently discards some data, so the original cannot be perfectly recovered. Uses (1): lossless for text/source code/program files (or PNG/ZIP), where every bit matters; lossy for photographs/audio/video (e.g. JPEG, MP3), where a small quality loss is acceptable in exchange for a much smaller file.

Total: 14 + 6 = 20 marks